Aryan Sidharth Munugoti

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Skills

Programming Languages and Database Management: Python, MySQL, HTML, CSS.
Data Processing and Libraries: NumPy, Pandas, Basics SQL, GitHub.
Data Analytics and Machine Learning: Data Cleaning and Preprocessing, Text Analysis, Machine Learning Basic.
AI/ML and Data Science: LLMs, Natural Language Processing (NLP), Topic Modeling (LDA), Neural Networks, CNN basics
Business Intelligence and Visualization: Excel, Power BI, Tally ERP, Matplotlib.
Soft Skills: Problem-solving, Effective Communication, Teamwork, Critical Thinking, Adaptability to New Technologies.
Project Methodology: Software Requirements Specification (SRS), Feature Design & Implementation
Other Skills: Eye for Detail, Time and Management, Corporate Social Responsibility, Event Management, Project Management, Image and Video Editing.

Education

Woxsen University 2022 – 2026
B.Tech in Computer Science and Engineering, Specialization in AI & ML

Harvest Public School
MPC, Central Board of Secondary Education 2020 - 2022
Gorkey Public School
Central Board of Secondary Education 2018 –2020

Projects

An Advanced Safety Helmet for Improved Two-Wheel Riders

- *The main purpose of this invention is to check and manage the condition of the user's helmet before and during driving. It uses different methods to examine the user's state, such as checking if they have consumed alcohol, if they are feeling stressed, if the helmet is properly worn, and other important factors. These checks are done to make sure the user and others around them are safe while driving.*
  - *The vehicle automatically stops by reducing the speed in steps (controlling the fuel to the engine control unit (ECU)) due to any one of the following factors:*
  - *If the helmet is displaced from its appropriate position.*
  - *If the rider is detected with alcohol consumption beyond the limit.*
  - *If the rider is diagnosed with higher stress levels.*
  - *If the rider is using the mobile by inserting the mobile phone into the helmet*

BHEEMA.AI

- *AI-Assisted Gym Training and Pose Analysis System.*
  ***Pose Correction System**: Developed a real-time pose correction system using Mediapipe and OpenCV to enhance exercise accuracy and safety.*
  ***Real-Time Feedback**: Implemented Google Text-to-Speech (gTTS) for AI-assisted voice feedback, delivering real-time posture guidance to users.*
  ***Exercise Recognition:** Built a system to automatically identify and count repetitions for biceps, triceps, and leg exercises.*
  ***User Interface**: Designed a user-friendly interface for displaying feedback, pose correction status and enhanced user engagement by creating an intuitive interface that supports diverse exercise categories (e.g., biceps, triceps, legs).*
  ***Privacy-First Design**: Prioritized privacy through secure handling of workout data and ensured compliance with modern data protection standards.*
  Delivered a system achieving 90% accuracy in pose recognition and repetition counting during functional testing.
  ***Tools & Technologies:** Python, OpenCV, Mediapipe, TensorFlow, Google Text-to-Speech (gTTS)*

IVR Automation Project - Dhvani.AI

- *Designed and implemented an NLP-powered Interactive Voice Response (IVR) system, integrating Transformer-based models for question-answering and natural language understanding. Developed a language detection model with support for diverse linguistic variations, improving query routing efficiency by 15%.*
- *Built a similar word prediction model enhancing user interaction by providing contextually accurate suggestions. Spearheaded the integration of the IVR system with real-time alerts, facial recognition modules, and alarm systems for end-to-end security and communication workflows.*
- *Conducted extensive data augmentation and preprocessing, using techniques like paraphrasing and back-translation to improve model performance by 20% on diverse datasets.*

  Tools & Technologies: *Hugging Face Transformers, Visual Studio Code, Amazon Web Services (AWS), BERT.*

AI ASSISTANT FOR CAMPUS LIFE

- *Developed a full-stack AI assistant web app using the MERN stack to help students manage schedules, events, and tasks with real-time chatbot support.*
- *Built REST APIs for syncing academic data from ERP system and encrypted data flow.*
- *Designed an interactive dashboard with React.js, showing task priorities, alerts, and robot support status.*

  Tools & Technologies*: OpenAI API, HuggingFace Transformers, MERN Stack, GPT, LLaMA2, Ollama*

SWAP: Smart Water Purification Bottle for Emergency Use

- *Social Innovation Project – SDG6 (Clean Water & Sanitation)*

  Designed a life-saving purification bottle (SWAP) to convert unhygienic water into safe drinking water using chlorination based on real-time pH sensor analysis. Engineered the device for high-impact use in disaster zones, survival scenarios, military operations, and public health emergencies.
- *Advocated for community awareness on hygiene and water safety through social media and university campaigns aligned with UN SDG 6.*
- *Collaborated with a cross-functional team on design thinking, prototyping, and user-centric development to ensure usability and field-readiness.*

  Tools & Technologies: *Water pH sensor, Microcontroller, Automated chlorination mechanism, 3D Design for Bottle Prototype*

Python Mini Projects

- *Developed Python-based mini projects including a Quiz Game and Number Guessing Game to demonstrate programming logic and user interaction and House Price Prediction using Linear Regression.*
- *Tools: Python*

Certifications and Achievements

- Submitted for Patent Rights: An Advanced Safety Helmet for Improved Two-Wheel Rider Safety.
- Machine Learning Workshop – IIT Hyderabad.
- 24 Hour National Level Hackathon – MGIT - Hyderabad.
- Smart India Hackathon - 2023
- *Web Development*
- Object-Oriented Hierarchies in Java – Coursera.
- Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning – Coursera.
- Introduction to Databases - Coursera Operating Systems Fundamentals – Coursera.